

Lecture 8

The classical r -matrix, the Sklyanin bracket and the CYBE

Objectives. *The deepest lecture in Part I, and the hinge of the whole course. [Lecture 7](#) built a machine that manufactures conserved quantities and then admitted it had not delivered what [Lecture 4](#) asked for: conservation is not involution. This lecture pays that debt. A single structural assumption about the Poisson brackets of a Lax matrix — that they are a commutator with one fixed numerical matrix r — turns out to imply involution of all the spectral invariants at once, and consistency of that assumption produces the classical Yang–Baxter equation. Part II is, in its entirety, the deformation of the equation proved in § 8.6.*

8.1 The question, and L7’s IOU

[Lecture 7](#) showed that a Lax pair $dL/dt = [M, L]$ makes the spectrum of L constant, so that every $\text{tr } L^k$ is a first integral. It also warned, in a box, that this is not enough. [Lecture 4](#) requires n integrals that are independent *and in involution*, and isospectrality delivers neither:

- *Independence* is a property of the particular model, checked by hand. [Lecture 9](#) does it for Toda.
- *Involution* — $\{\text{tr } L^n, \text{tr } L^m\} = 0$ — does not follow from $dL/dt = [M, L]$ at all. That equation says the $\text{tr } L^k$ are constant along the flow of H ; it says nothing whatever about their brackets with each other.

That is the IOU. Paying it needs new information, and the information cannot be about the *dynamics*, since involution is a statement about the phase-space brackets of L with itself, made without reference to any Hamiltonian. So the question is:

what does one have to assume about $\{L_{ab}, L_{cd}\}$ to force all the $\text{tr } L^k$ into involution?

The answer, found by Sklyanin, is startlingly cheap, and stating it requires one piece of notation that is worth twenty minutes on its own.

8.2 Tensor legs: the notation, computed twice

Reading

The tensor product itself is a prerequisite for this course (*notes/part-I-syllabus.md* §2), at the level of two-spin systems in quantum mechanics. What follows is not the tensor product; it is a compression scheme for writing large collections of Poisson brackets, and it is the compression, not the algebra, that students find hard.

The problem the notation solves

L is a $d \times d$ matrix whose entries are functions on phase space. The object we need to talk about is the collection of *all* the brackets

$$\{L_{ab}, L_{cd}\}, \quad a, b, c, d = 1, \dots, d, \quad (8.1)$$

which for $d = 2$ is sixteen numbers and for $d = 3$ is eighty-one. Writing them one at a time is hopeless. The trick is to package them as a *single matrix*, of size $d^2 \times d^2$, acting on $V \otimes V$ with $V = \mathbb{C}^d$.

Definition 8.1 (Tensor legs). Write E_{ab} for the matrix unit with a 1 in row a , column b , and recall $(A \otimes B)_{(ac),(bd)} = A_{ab}B_{cd}$, the row labelled by the pair (a, c) and the column by (b, d) . For a $d \times d$ matrix L put

$$L_1 = L \otimes \mathbb{1}, \quad L_2 = \mathbb{1} \otimes L, \quad (8.2)$$

and define the $d^2 \times d^2$ matrix

$$\{L_1, L_2\} = \sum_{a,b,c,d} \{L_{ab}, L_{cd}\} E_{ab} \otimes E_{cd}, \quad \text{that is} \quad \left(\{L_1, L_2\} \right)_{(ac),(bd)} = \{L_{ab}, L_{cd}\}. \quad (8.3)$$

The subscript is a *leg*, not a matrix index: L_1 is the whole matrix L , placed in the first factor. The right-hand equality in (8.3) is the one to memorise — the entry in row (a, c) and column (b, d) is the bracket of the (a, b) entry with the (c, d) entry.



Figure 8.1: Tensor legs. Each horizontal line is a copy of V ; a box sitting on a line is a matrix acting in that factor and doing nothing to the others. The subscript names the line, not a matrix entry.

Remark 8.2 (The confusion to head off now). L_1 and L_2 **commute as matrices**: $(L \otimes \mathbb{1})(\mathbb{1} \otimes L) = L \otimes L = (\mathbb{1} \otimes L)(L \otimes \mathbb{1})$, because they act in different factors. Their **entries do not Poisson-commute**: that is exactly what (8.3) records, and if it were zero this lecture would have no subject. Two different brackets are in play — the matrix commutator and the Poisson bracket — and [Lecture 1](#) warned that $[\cdot, \cdot]$ carries both meanings. Here they sit in the same formula. *The involution theorem of § 8.4 and the composition argument of § 8.8 both turn on $[L_1, L_2] = 0$ at their decisive step, and nothing in this lecture uses $\{L_1, L_2\} = 0$, which is false.*

The same bracket, computed twice

Take the harmonic oscillator's Lax matrix from [Lecture 7](#), with one degree of freedom and $\{q, p\} = 1$:

$$L = \begin{pmatrix} p & q \\ q & -p \end{pmatrix}, \quad \text{tr } L^2 = 2(p^2 + q^2) = 4H. \quad (8.4)$$

Compactly. The brackets of the entries are $\{p, q\} = -1$, $\{q, p\} = +1$, and everything else built from these by linearity. Feeding them into the sum in (8.3),

$$\{L_1, L_2\} = -E_{11} \otimes E_{12} - E_{11} \otimes E_{21} + E_{12} \otimes E_{11} - E_{12} \otimes E_{22} + E_{21} \otimes E_{11} - E_{21} \otimes E_{22} + E_{22} \otimes E_{12} + E_{22} \otimes E_{21}.$$

Explicitly. The same object as a 4×4 matrix, rows and columns ordered $(1, 1), (1, 2), (2, 1), (2, 2)$, each entry read off directly from the right-hand form of (8.3):

$$\{L_1, L_2\} = \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{pmatrix}. \quad (8.5)$$

Check one entry against the other form before reading on. Row $(1, 2)$, column $(2, 2)$ means $a = 1, c = 2, b = 2, d = 2$, so the entry is $\{L_{12}, L_{22}\} = \{q, -p\} = -1$; and in the compact form the term $-E_{12} \otimes E_{22}$ is the only one contributing there. *If this paragraph is not comfortable, nothing after it will be.* Sheet 8 drills it.

The permutation operator

One particular matrix on $V \otimes V$ appears everywhere below.

Definition 8.3 (Permutation operator). $P(v \otimes w) = w \otimes v$; equivalently $P = \sum_{a,b} E_{ab} \otimes E_{ba}$. For $d = 2$,

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Proposition 8.4 (Properties of P). $P^2 = \mathbb{1}$; $P(A \otimes B)P = B \otimes A$; and

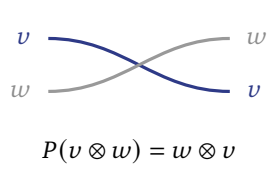
$$(E_{ab} \otimes E_{cd}) P = E_{ad} \otimes E_{cb}. \quad (8.6)$$

Proof. P^2 swaps twice. For the third, expand P and use $E_{ab}E_{ef} = \delta_{be}E_{af}$:

$$(E_{ab} \otimes E_{cd}) \sum_{e,f} E_{ef} \otimes E_{fe} = \sum_{e,f} \delta_{be} E_{af} \otimes \delta_{df} E_{ce} = E_{ad} \otimes E_{cb},$$

the sum collapsing at $e = b, f = d$. The second property follows by applying both sides to $v \otimes w$. ■

8.3 The Sklyanin bracket



P is a crossing, and $P^2 = \mathbb{1}$ says that uncrossing twice does nothing. Every diagram in [Lecture 17](#) is built from this one picture; here it is only bookkeeping.

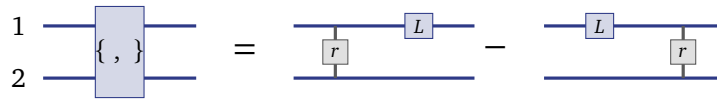
Figure 8.2: The permutation operator drawn as a crossing of the two legs.

Definition 8.5 (Sklyanin bracket). A Lax matrix $L(\lambda)$ satisfies a *Sklyanin bracket* with classical r -matrix r if

$$\{L_1(\lambda), L_2(\mu)\} = [r_{12}(\lambda - \mu), L_1(\lambda) + L_2(\mu)], \quad (8.7)$$

where $r_{12}(\lambda)$ is a matrix on $V \otimes V$ of *numbers* — it does not depend on the point of phase space.

Read what (8.7) asserts. On the left is an arbitrary $d^2 \times d^2$ matrix of Poisson brackets, d^4 functions on phase space with no reason to be related. On the right is a commutator of one fixed numerical matrix with something built linearly from L itself. The assumption is enormously restrictive, and its whole force is that *the same r works at every point of phase space and for every pair λ, μ .*



r is a *rung* joining the two legs; the commutator is the difference of the two orders.

Figure 8.3: The Sklyanin bracket (8.7), drawn for the L_1 half of $L_1 + L_2$; the L_2 half is the same picture with the box on the lower leg. The left-hand side is an unknown matrix of Poisson brackets; the right-hand side says it equals one fixed rung r slid past L , minus L slid past the rung. Note that the two terms are added, not multiplied — the picture with an L on *both* legs is the *quadratic* bracket of § 8.8, which is a different relation.

Two qualifications, both of which the course later cashes:

- *Numerical.* If r is allowed to depend on the dynamical variables the bracket is called *dynamical*, and [Lecture 11](#) meets the first real example — Calogero–Moser has no r of the constant form. Everything proved in § 8.4 survives that generalisation; § 8.6 does not.
- *Which r ?* This lecture teaches how to *read and check* an r -matrix, never how to solve for one. Finding r from scratch is outside Part I, and Sheet 8 accordingly supplies a candidate and asks you to verify it.

Remark 8.6 (r is not arbitrary). Antisymmetry of the Poisson bracket already constrains r . Reading (8.3) backwards, $\{L_{ab}, L_{cd}\} = -\{L_{cd}, L_{ab}\}$ is the matrix statement $\{L_1(\lambda), L_2(\mu)\} = -P \{L_1(\mu), L_2(\lambda)\} P$. Substituting (8.7) on both sides and writing $r_{21} = P r_{12} P$, the two agree as soon as

$$r_{12}(\lambda) = -r_{21}(-\lambda). \quad (8.8)$$

The rational solution of § 8.7 satisfies it, since $PPP = P$ makes $r_{21} = r_{12}$ there, and $r(\lambda) = cP/\lambda$ is odd. § 8.6 uses (8.8). *This is core material, not an aside:* the proof of Theorem 8.11 uses (8.8) to trade r_{21} for $-r_{12}$, and without it that proof does not start.

8.4 The involution theorem

Theorem 8.7 (Involution from the Sklyanin bracket). *If $L(\lambda)$ satisfies (8.7), then for all $n, m \geq 1$ and all λ, μ at which it holds,*

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = 0. \quad (8.9)$$

The qualification is needed because the r -matrices that occur have a pole at coincident arguments — the rational one of § 8.7 is $cP/(\lambda - \mu)$ — so (8.7) is an assumption about $\lambda \neq \mu$ and says nothing at $\lambda = \mu$. Nothing is lost: the left of (8.9) is continuous in λ and μ whenever L is, and a continuous function vanishing off the diagonal vanishes on it too.

The proof is three steps and each is short. It is the most important argument in Part I.

Proof. Step 1: the chain rule, in tensor-leg form. Both $\mathrm{tr} L(\lambda)^n$ and $\mathrm{tr} L(\mu)^m$ are functions of the entries of L , so by bilinearity and the Leibniz rule of Lecture 1,

$$\{F, G\} = \sum_{a,b,c,d} \frac{\partial F}{\partial L_{ab}} \frac{\partial G}{\partial L_{cd}} \{L_{ab}, L_{cd}\}.$$

For $F = \mathrm{tr} L^n$ we have $\partial F / \partial L_{ab} = n (L^{n-1})_{ba}$, and similarly for G . Comparing with (8.3), the double sum is exactly a trace over both legs:

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = nm \mathrm{tr}_{12} \left(L_1(\lambda)^{n-1} L_2(\mu)^{m-1} \{L_1(\lambda), L_2(\mu)\} \right). \quad (8.10)$$

Step 2: substitute the assumption. Put $A = L_1^{n-1} L_2^{m-1}$ and $B = L_1 + L_2$. Then (8.7) turns (8.10) into

$$\{\mathrm{tr} L(\lambda)^n, \mathrm{tr} L(\mu)^m\} = nm \mathrm{tr} (A [r, B]).$$

Step 3: the trace of that vanishes. Here is the only place any structure is used, and it is Remark 8.2: L_1 and L_2 commute as matrices, so A — a product of their powers — commutes with B . Hence, using cyclicity of the trace twice and $AB = BA$ once,

$$\mathrm{tr} (A [r, B]) = \mathrm{tr} (ArB) - \mathrm{tr} (ABr) = \mathrm{tr} (ArB) - \mathrm{tr} (rAB) = \mathrm{tr} (ArB) - \mathrm{tr} (rBA) = \mathrm{tr} (ArB) - \mathrm{tr} (ArB) = 0. \quad \blacksquare$$

Lecture 7's IOU is paid. Note how little was spent: the argument never opened r , never used the Lax equation, and never mentioned a Hamiltonian. It also never used the *particular* combination $L_1 + L_2$ — only that it commutes with A . That observation costs nothing now and is exactly what § 8.8 will need.

Corollary 8.8 (What the model gets). *Expanding $\mathrm{tr} L(\lambda)^k$ in powers of λ produces a family of functions in involution, and any Hamiltonian among them has all the others as commuting integrals. Whether there are n independent ones is a separate question about the model, settled for Toda in Lecture 9.*

8.5 The converse: Babelon–Viallet

Theorem 8.7 is one-way: a Sklyanin bracket gives involution. The converse is true but weaker than one would like.

Theorem 8.9 (Babelon–Viallet · cited). *If the eigenvalues of L are in involution on a neighbourhood where L is regular and semisimple, then L satisfies*

$$\{L_1, L_2\} = [r_{12}, L_1] - [r_{21}, L_2], \quad (8.11)$$

with r in general dynamical — a function on phase space.

Read (8.11) carefully: it is not (8.7). Two independent matrices appear, and the single-commutator form is the special case $r_{21} = -r_{12}$. The converse does *not* deliver that special case, and the difference is not a technicality — it is the difference between a true statement and a false one.

Example 8.10 (Why the strengthening fails). On a canonical (q, p) plane put $D = \text{diag}(a, b)$ with $a \neq b$, and

$$L(q, p) = UDU^{-1}, \quad U = e^{qE_{12}}e^{pE_{21}}.$$

Then L is regular semisimple near the origin and its eigenvalues are the *constants* a, b , so they are in involution for the most trivial of reasons. At the origin $\partial_q L = [E_{12}, D] = (b - a)E_{12}$ and $\partial_p L = [E_{21}, D] = (a - b)E_{21}$, so

$$\{L_{12}, L_{21}\} \Big|_0 = -(a - b)^2 \neq 0.$$

Now look at where that number would have to come from in (8.7). At the origin $L_1 + L_2 = D \otimes \mathbb{1} + \mathbb{1} \otimes D$ is diagonal with entries $2a, a + b, a + b, 2b$, and the required entry sits between the two *equal* ones. For any matrix r whatever, dynamical or not, a commutator with a diagonal matrix has that entry multiplied by the difference of the two diagonal entries — here zero. So no r can satisfy (8.7) for this L , while (8.11), with its two independent terms, has no such obstruction.

So an r -matrix always exists behind involution, but it need not be numerical and it need not put the bracket in the single-commutator form of Definition 8.5. The classification of § 8.7 applies to neither. Lecture 11 meets a dynamical r in Calogero–Moser, so the caveat is not hypothetical.

8.6 The classical Yang–Baxter equation

The Sklyanin bracket is an assumption about a Poisson bracket, so it must be consistent with the Jacobi identity. Working out what that requires produces the equation the rest of the course is about.

Put three legs in play, $V \otimes V \otimes V$, write L_i for L in leg i at spectral parameter λ_i , and abbreviate $r_{ij} = r_{ij}(\lambda_i - \lambda_j)$. Two facts do the work: $[r_{ij}, L_k] = 0$ when $k \notin \{i, j\}$, because they act on disjoint legs; and $r_{21} = -r_{12}$, which Remark 8.6 showed to be *sufficient* for antisymmetry of the Poisson bracket but which the bracket does not by itself force. It is

therefore a hypothesis of [Theorem 8.11](#), not a consequence of [Definition 8.5](#), and is assumed from here on.

Theorem 8.11 (Jacobi and the CYBE). *For a Lax matrix satisfying (8.7) with an r obeying the skew condition (8.8), the Jacobi identity for L_1, L_2, L_3 is equivalent to*

$$[[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}], L_1 + L_2 + L_3] = 0. \quad (8.12)$$

In particular it holds whenever r satisfies the classical Yang–Baxter equation

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0. \quad (8.13)$$

Proof. Since r is numerical it passes through a Poisson bracket, so

$$\{\{L_1, L_2\}, L_3\} = [r_{12}, \{L_1, L_3\} + \{L_2, L_3\}] = [r_{12}, [r_{13}, L_1 + L_3] + [r_{23}, L_2 + L_3]].$$

Use $[r_{12}, L_3] = 0$ and the *matrix* Jacobi identity in the form $[A, [B, C]] = [[A, B], C] + [B, [A, C]]$ on each of the two terms. The first gives

$$[r_{12}, [r_{13}, L_1 + L_3]] = [[r_{12}, r_{13}], L_1 + L_3] + [r_{13}, [r_{12}, L_1]],$$

and the second the same with (13) $\rightarrow$ (23), $L_1 \rightarrow L_2$. Now add the two cyclic permutations of this, replacing r_{21}, r_{31}, r_{32} by $-r_{12}, -r_{13}, -r_{23}$ as (8.8) allows. The six leftover double commutators pair up, again by matrix Jacobi: for instance

$$[r_{13}, [r_{12}, L_1]] - [r_{12}, [r_{13}, L_1]] = -[[r_{12}, r_{13}], L_1].$$

Now collect. The first group contributes each double commutator of r 's twice, once from each of the two cyclic terms that produce it, and the two contributions carry different pairs of L_i :

$$[r_{12}, r_{13}] : L_1 + L_2 \text{ and } L_1 + L_3; \quad [r_{12}, r_{23}] : L_1 + L_2 \text{ and } L_2 + L_3; \quad [r_{13}, r_{23}] : L_1 + L_3 \text{ and } L_2 + L_3,$$

so their totals are $2L_1 + L_2 + L_3$, $L_1 + 2L_2 + L_3$ and $L_1 + L_2 + 2L_3$ respectively. The three surviving pairs of double commutators, computed as displayed above, contribute exactly $-[[r_{12}, r_{13}], L_1]$, $-[[r_{12}, r_{23}], L_2]$ and $-[[r_{13}, r_{23}], L_3]$, which remove the one doubled term in each line. All three are left with $L_1 + L_2 + L_3$, and that is (8.12). ■

Remark 8.12 (Machine-checked). The collection above is exactly the kind of bookkeeping in which a sign is lost. It is verified in `scripts/check-l8-algebra.py`, for $d = 2$, with three explicit legs and *symbolic* entries throughout — sixty free parameters, so the result is an identity and not a sample. The Jacobi sum equals $[X, L_1 + L_2 + L_3]$, and — the point of [Remark 8.13](#) — is *not* identically zero on its own.

Remark 8.13 (Jacobi does not force the CYBE, and saying so matters). [Theorem 8.11](#) gives (8.12), and (8.13) is *sufficient* for it, not implied by it: the Jacobi identity only requires the triple sum to *commute with* $L_1 + L_2 + L_3$, and for a particular L that is a weaker demand. What makes (8.13) the right thing to impose is that it mentions no L at all. One solution r then serves every model whose Lax matrix it fits, which is why the classification of r -matrices is a subject and the classification of integrable models is not.

One variant deserves its name in passing. Replacing the zero on the right of (8.13) by a specific r -independent term gives the *modified classical Yang–Baxter equation*, whose solutions generate the same structures over a slightly larger class of Lie algebras. Its theory belongs to Lie-bialgebra and Poisson–Lie theory, which this course does not enter; it is named here so that meeting the phrase elsewhere is not a surprise.

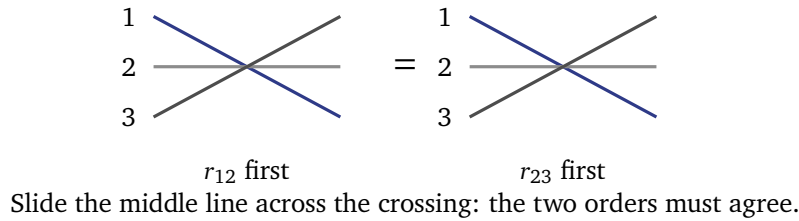


Figure 8.4: The classical Yang–Baxter equation. Three lines cross pairwise, and the two ways of resolving the triple crossing — sliding line 2 past the crossing of 1 and 3 either before or after — give the same answer. This is the picture [Lecture 0](#) promised and the one [Lecture 17](#) quantises.

8.7 The rational solution

Proposition 8.14 (The rational r -matrix). $r(\lambda) = cP/\lambda$ satisfies the CYBE (8.13) for every constant c .

Proof. Write $\lambda_{ij} = \lambda_i - \lambda_j$. The permutation operators satisfy $P_{12}P_{13} = P_{13}P_{23} = P_{23}P_{12}$ and $P_{13}P_{12} = P_{12}P_{23} = P_{23}P_{13}$ — both are the two ways of cycling three legs. Calling the difference of those two products K , all three commutators are $\pm K$:

$$[P_{12}, P_{13}] = K, \quad [P_{12}, P_{23}] = -K, \quad [P_{13}, P_{23}] = K.$$

So the left of (8.13) is

$$c^2 K \left(\frac{1}{\lambda_{12}\lambda_{13}} - \frac{1}{\lambda_{12}\lambda_{23}} + \frac{1}{\lambda_{13}\lambda_{23}} \right) = c^2 K \frac{\lambda_{23} - \lambda_{13} + \lambda_{12}}{\lambda_{12}\lambda_{13}\lambda_{23}} = 0,$$

because $\lambda_{12} + \lambda_{23} = \lambda_{13}$ identically. ■

The cancellation is a statement about the three spectral parameters and nothing else, which is why c is free: the normalisation of r is fixed by the Lax matrix it is paired with, not by the CYBE. [Lecture 10](#) fixes it for the periodic Toda chain; the open chain of [Lecture 9](#) needs a different, constant r .

Reading

Two further families solve (8.13): the *trigonometric* r -matrices, in which $1/\lambda$ is replaced by $\cot \lambda$ or $\coth \lambda$ and the permutation is deformed, and the *elliptic* ones, built from Weierstrass functions. Belavin and Drinfeld classified the solutions *under hypotheses this course has not set up*: for r valued in $\mathfrak{g} \otimes \mathfrak{g}$ with $\mathfrak{g}$ a finite-dimensional *simple* Lie algebra, in difference form, and non-degenerate in their technical sense (the residue at the simple pole is proportional to the Casimir of an invariant form), every solution is equivalent to a rational, trigonometric or elliptic one *· cited*, and the three degenerate into one another in limits. Those hypotheses are doing real work — dropped, the statement is false, and already fails for $d = 1$, where every commutator in (8.13) vanishes and any function of λ with a simple pole at the origin solves it. The same trichotomy will reappear in [Lecture 11](#) as the four Calogero–Moser potentials, and in Part II as XXX, XXZ and XYZ — it is the same list every time, which is not a coincidence.

8.8 Monodromy and transfer matrices

Everything so far concerned a single Lax matrix. A lattice model has one per site, and the object that organises them is a product.

Definition 8.15 (Classical monodromy and transfer matrices). For a chain of N sites with Lax matrices $L^{(n)}(\lambda)$, the *monodromy matrix* and *transfer matrix* are

$$T(\lambda) = L^{(N)}(\lambda) \cdots L^{(1)}(\lambda), \quad t(\lambda) = \text{tr } T(\lambda). \quad (8.14)$$

The chain is *ultralocal* if variables at different sites Poisson-commute, $\{L_{ab}^{(n)}, L_{cd}^{(m)}\} = 0$ for $n \neq m$.

Remark 8.16 (A naming collision, and it is only that). The *monodromy matrix* $T(\lambda)$ has nothing to do with the *Hamiltonian monodromy* of [Lecture 4](#) — the global twisting of a torus bundle. Both words come from “going around”: there, a loop in the space of regular values; here, a circuit of the lattice. There the relationship ends, and a student who met the word in [Lecture 4](#) should expect no connection, because there is none.

Now a point that is easy to slide past. The bracket (8.7) is *linear* in L on the right-hand side, and that form does *not* survive multiplication: computing $\{T_1, T_2\}$ for a product of two Lax matrices does not return $[r, T_1 + T_2]$. The form that does survive is the *quadratic* one.

Theorem 8.17 (The quadratic bracket composes). Suppose the chain is ultralocal and each site satisfies

$$\{L_1^{(n)}(\lambda), L_2^{(n)}(\mu)\} = [r_{12}(\lambda - \mu), L_1^{(n)}(\lambda) L_2^{(n)}(\mu)]. \quad (8.15)$$

Then T satisfies the same relation, $\{T_1, T_2\} = [r_{12}, T_1 T_2]$.

Proof. Take $N = 2$; the general case is the same telescoping. Leibniz plus ultralocality kills the cross terms, and matrices in different legs commute, so with $A = L_1^{(2)} L_2^{(2)}$ and $B = L_1^{(1)} L_2^{(1)}$,

$$\{T_1, T_2\} = [r, A] B + A [r, B] = rAB - ArB + ArB - ABr = [r, AB],$$

and $AB = T_1 T_2$ because leg-1 and leg-2 matrices commute. ■

Corollary 8.18 (Commuting transfer matrices). $\{t(\lambda), t(\mu)\} = 0$ for all λ, μ .

Proof. $\{t(\lambda), t(\mu)\} = \text{tr}_{12} \{T_1, T_2\} = \text{tr}_{12} [r, T_1 T_2] = 0$: the trace of a commutator vanishes. The same one-line argument works for the linear bracket (8.7), and [Theorem 8.7](#) likewise needs only that its B commute with A — which $T_1 T_2$ does. So both forms deliver involution; only the quadratic one composes. ■

So $t(\lambda)$ is a generating function for commuting charges: expand it in λ and every coefficient Poisson-commutes with every other. [Lecture 10](#) builds exactly this for the periodic Toda chain, and [Lecture 17](#) builds its quantum counterpart, where the same expansion produces the Hamiltonian of the Heisenberg spin chain.

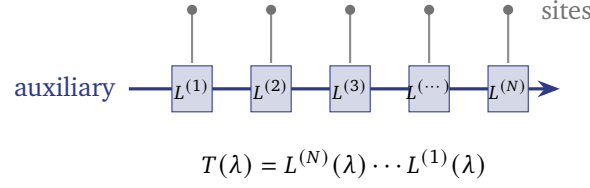


Figure 8.5: The monodromy matrix as transport along the lattice. One auxiliary line threads every site in turn, picking up each site's Lax matrix; the vertical stubs are the physical sites, which carry the dynamical variables.

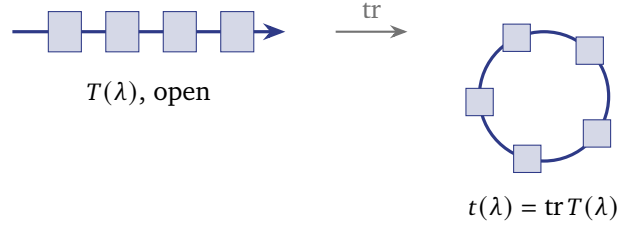


Figure 8.6: Taking the trace closes the auxiliary line into a loop. The loop has no ends to distinguish, which is the picture behind cyclicity — and cyclicity is the whole proof of [Corollary 8.18](#).

8.9 The dictionary, and $R = \mathbb{1} + i\hbar r + O(\hbar^2)$

Everything in this lecture has a quantum counterpart, and Part II is the systematic construction of them. The table is worth seeing before the details.

Part I (classical)	Part II (quantum)
Poisson bracket $\{\cdot, \cdot\}$	commutator $[\cdot, \cdot]$
Lax matrix $L(\lambda)$	Lax operator $L_n(\lambda)$
monodromy matrix $T(\lambda)$	monodromy matrix $T(\lambda)$
$t = \text{tr } T$, $\{t(\lambda), t(\mu)\} = 0$	$t = \text{tr}_a T$, $[t(\lambda), t(\mu)] = 0$
quadratic bracket (8.15)	the RTT relation
classical r -matrix $r(\lambda)$	quantum R -matrix $R(\lambda)$
CYBE (8.13)	Yang–Baxter equation

The link between the last two rows is the expansion frozen for this course,

$$R(\lambda) = \mathbb{1} + i\hbar r(\lambda) + O(\hbar^2). \quad (8.16)$$

Put (8.16) into the quantum relation, use $T_1 T_2 - T_2 T_1 = i\hbar \{T_1, T_2\} + O(\hbar^2)$, and the terms of order $\hbar$ reproduce [Theorem 8.17](#) exactly — *provided the two monodromies are ordered as*

$$R_{12}(\lambda - \mu) T_2(\mu) T_1(\lambda) = T_1(\lambda) T_2(\mu) R_{12}(\lambda - \mu). \quad (8.17)$$

Remark 8.19 (A convention, pinned [here](#)). The literature writes (8.17) with the two sides exchanged as often as not, and the two orderings correspond to $R = \mathbb{1} \pm i\hbar r$. They are not both available to us: (8.16) is frozen with a $+$, and that choice *forces* the ordering in (8.17). Taking the other ordering with a $+$ returns $\{T_1, T_2\} = -[r, T_1 T_2]$, the Sklyanin bracket with $r \rightarrow -r$. [Lecture 17](#) does this limit properly; what matters here is that the dictionary is *forced*, not chosen, once the classical side is fixed.

Summary.

- *Tensor legs.* $\{L_1, L_2\}$ is the $d^2 \times d^2$ matrix whose $(ac), (bd)$ entry is $\{L_{ab}, L_{cd}\}$ (Definition 8.1). L_1 and L_2 commute as matrices while their entries do not Poisson-commute — Remark 8.2, and every proof here turns on it.
- *The Sklyanin bracket.* $\{L_1, L_2\} = [r_{12}, L_1 + L_2]$ with r numerical (Definition 8.5).
- *Involution.* It implies $\{\text{tr } L(\lambda)^n, \text{tr } L(\mu)^m\} = 0$ (Theorem 8.7), in three lines, using nothing about r . This is Lecture 7's IOU, paid.
- *The CYBE.* Jacobi forces (8.12); the CYBE (8.13) is the L -independent sufficient condition one imposes (Remark 8.13), and $r = cP/\lambda$ solves it (Proposition 8.14).
- *Transfer matrices.* The quadratic bracket, not the linear one, survives multiplication (Theorem 8.17), and hence $\{t(\lambda), t(\mu)\} = 0$ (Corollary 8.18) — a generating function for commuting charges.
- *The bridge.* $R = \mathbb{1} + i\hbar r + O(\hbar^2)$, and that sign fixes the ordering in the quantum relation (8.17).

Tutorial. Exercise Sheet 8.

References

Babelon, Bernard and Talon, *Introduction to Classical Integrable Systems*, Ch. 2–3 (r -matrices, the Sklyanin bracket, involution); Faddeev and Takhtajan, *Hamiltonian Methods in the Theory of Solitons*, Part I Ch. 1 (the classical r -matrix and the transition to the quantum case); Arutyunov, *Elements of Classical and Quantum Integrable Systems*, Ch. 1 (Babelon–Viallet, dynamical r -matrices). **Conventions for the placement of T_1, T_2 and for the sign in (8.16) differ between all three** — see Remark 8.19.